

Ajay Alle

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SUMMARY

Final-year Electronics & Communication Engineering student with a strong foundation in Python, data analysis, and computer vision, building toward Data/AI-ML engineering roles. Delivered a real-time computer-vision gesture-recognition pipeline (OpenCV, MediaPipe) as part of a embedded systems team, and an independent IoT attendance system (ESP32) that automated classroom record-keeping end to end. Comfortable with NumPy, Pandas, scikit-learn, and SQL, with exposure to TensorFlow/PyTorch. Strong hands-on builder mindset backed by certifications in AI & Green Skills and Python-based computing.

PROJECTS

Gesture-Controlled Smart Home Automation System (2026) — *Python, OpenCV, MediaPipe, Raspberry Pi, Arduino, Serial Communication*

Built an internet-free home automation system as part of a 6-member team, controlling appliances via real-time hand-gesture recognition instead of Wi-Fi or cloud dependency. Owned the computer-vision pipeline: developed gesture detection using OpenCV and MediaPipe on a Raspberry Pi, mapping finger-count patterns to appliance control commands, and interfaced it with an Arduino controller over serial communication to drive relay-based switching. Designed a triple-layer control system (gesture, IR remote, manual switch) with priority logic so manual or IR input could safely override gesture commands, and tested detection accuracy across varied lighting and hand positions to improve pipeline reliability.

Wi-Fi Based Attendance System Using MAC Address Detection (2025) — *ESP32, Python, Google Apps Script, Google Sheets*

Automated student attendance tracking by detecting connected devices' MAC addresses over Wi-Fi via an ESP32 microcontroller, removing manual rollcall. Built a Google Apps Script pipeline to log detections directly into Google Sheets, auto-calculating attendance percentage and flagging low-attendance students, with an LCD and physical-button interface for subject/semester selection supporting multi-device connectivity in real time. Delivered a scalable, paperless attendance solution that reduced manual errors in classroom record-keeping.

TECHNICAL SKILLS

Languages: Python, C, C++

Data & ML: NumPy, Pandas

Computer Vision & Embedded: OpenCV, MediaPipe, ESP32, Arduino, IoT concepts, sensor & serial interfacing

Tools: Jupyter Notebook, Git basics, Google Apps Script

CERTIFICATIONS

- Edunet Skills4Future — Foundation Course on Artificial Intelligence and Green Skills
- Microsoft — Azure Cloud, AI/ML
- NPTEL (Swayam) — The Joy of Computing Using Python
- Outskill — Generative AI Mastermind
- Bootcamp — Drone Technology: Foundation and Applications

EDUCATION

Dr. S. & S. S. Ghandhy Government Engineering College Bachelor of Engineering in Electronics & Communication Engineering Current CGPA 7.52/10	2024 – 2027
Dr. S. & S. S. Ghandhy College of Engineering & Technology Diploma in Electrical Engineering CGPA: 6.90/10	2020 – 2023